Package ‘RCamelsPE’

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Type Package

Title Tools for Reading, Processing and Visualizing CAMELS-PE Data

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Description Provides tools to efficiently read, process and visualize hydrological, climatic, physiographic and geospatial data from the CAMELS-PE dataset (Catchment Attributes and Meteorology for Large-sample Studies - Peru). The package does not include the dataset and is designed to work with externally stored CAMELS-PE data. Functions support reading time series, catchment attributes, metadata, and geospatial data, as well as basic visualization of time series and spatial patterns.

License GPL (>= 2)

Encoding UTF-8

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URL https://github.com/yourusername/RCamelsPE

BugReports https://github.com/yourusername/RCamelsPE/issues

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Contents

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get_camels_path — *Get CAMELS-PE dataset path*

Description

Get CAMELS-PE dataset path

Usage

```
get_camels_path()
```

Value

Character path to the CAMELS-PE root directory.

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plot_attribute_map — *Plot CAMELS-PE Attribute Map*

Description

Creates a thematic map of CAMELS-PE catchments using a selected attribute. The function joins attribute data with catchment geometries by `gauge_id` and returns a `ggplot` object.

Usage

```
plot_attribute_map(
  catchments,
  attributes,
  variable,
  gauges = NULL,
  na_color = "grey80",
  ...
)
```

Arguments

`catchments`	An `sf` object with catchment polygons.
`attributes`	A data frame or tibble with CAMELS-PE attributes.
`variable`	Character string. Name of the attribute to visualize.
`gauges`	Optional `sf` object with gauge point locations.
`na_color`	Character string. Color for missing values.
`...`	Additional arguments passed to `ggplot2::geom_sf()` for catchments.

Value

A `ggplot` object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")
attr <- read_attributes("topographic")

plot_attribute_map(
  catchments = catchments,
  attributes = attr,
  variable = "area_km2",
  gauges = gauges
)

## End(Not run)
```

`plot_catchments` *Plot CAMELS-PE Catchments*

Description

Creates a simple map of CAMELS-PE catchments and, optionally, gauge locations. This function is useful for quick inspection of the spatial distribution of catchments and stations.

Usage

```
plot_catchments(catchments, gauges = NULL, fill = NULL, ...)
```

Arguments

`catchments`	An `sf` object with catchment polygons, typically obtained with `read_geospatial("catchments")`.
`gauges`	Optional `sf` object with gauge point locations, typically obtained with `read_geospatial("gauges")`
`fill`	Optional character string. Name of a catchment column used to fill polygons. If NULL, catchments are plotted with a constant fill.
`...`	Additional arguments passed to `ggplot2::geom_sf()` for catchments.

Value

A ggplot object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")

plot_catchments(catchments)
plot_catchments(catchments, gauges)

## End(Not run)
```

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plot_timeseries *Plot CAMELS-PE Time Series*

Description

Creates a time series plot for one CAMELS-PE variable, such as precipitation, observed streamflow, or temperature. The function returns a ggplot object, so it can be further customized using standard ggplot2 layers.

Usage

```
plot_timeseries(
  data,
  variable = "flow_obs",
  gauge_id = NULL,
  date_col = "date",
  facet = TRUE,
  scales = "free_y",
  ...
)
```

Arguments

data	A data frame or tibble containing CAMELS-PE time series.
variable	Character string. Name of the variable to plot.
gauge_id	Optional character vector. Gauge IDs to filter before plotting.
date_col	Character string. Name of the date column.
facet	Logical. If TRUE, creates one panel per gauge ID.
scales	Character string. Scales passed to facet_wrap().
...	Additional arguments passed to ggplot2::geom_line().

Value

A ggplot object.

Examples

```
## Not run:
ts <- read_timeseries(
  gauge_id = c("PE_0001", "PE_0002"),
  vars = c("date", "gauge_id", "prec", "flow_obs", "tmean")
)

plot_timeseries(ts, variable = "flow_obs")
plot_timeseries(ts, variable = "prec", linewidth = 0.4)
plot_timeseries(ts, variable = "flow_obs", facet = FALSE)

## End(Not run)
```

`read_attributes` *Read CAMELS-PE Catchment Attributes*

Description

Reads catchment attributes from the CAMELS-PE dataset. Attributes are stored by thematic groups, such as topography, climate, hydrological signatures, land cover, geology, soil, and human intervention.

Usage

```
read_attributes(type = "all", gauge_id = NULL, path = get_camels_path())
```

Arguments

`type`	Character string. Attribute group to read. One of `"topographic"`, `"climatic"`, `"hydrological"`, `"landcover"`, `"geologic"`, `"soil"`, `"human_intervention"`, or `"all"`.
`gauge_id`	Optional character vector. Gauge IDs used to filter the returned attributes.
`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with `set_camels_path()` is used.

Details

Available attribute types are:

- `"topographic"`: topographic attributes.
- `"climatic"`: climatic indices.
- `"hydrological"`: hydrological signatures.
- `"landcover"`: land-cover attributes.
- `"geologic"`: geologic attributes.
- `"soil"`: soil attributes.
- `"human_intervention"`: human intervention attributes.
- `"all"`: all available attribute groups joined by `gauge_id`.

Value

A tibble with CAMELS-PE catchment attributes.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read topographic attributes
topo <- read_attributes(type = "topographic")

# Read climatic attributes for selected catchments
clim <- read_attributes(
  type = "climatic",
  gauge_id = c("PE_0001", "PE_0002")
)

# Read all attributes
attr_all <- read_attributes(type = "all")

## End(Not run)
```

`read_geospatial` *Read CAMELS-PE Geospatial Data*

Description

Reads geospatial data from the CAMELS-PE dataset, including gauge locations and catchment boundaries. The data are stored as GeoPackage files in the `04_geospatial` directory.

Usage

```
read_geospatial(type = c("gauges", "catchments"), path = get_camels_path())
```

Arguments

`type`	Character string. Either `"gauges"` or `"catchments"`.
`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with `set_camels_path()` is used.

Details

Available types:

- `"gauges"`: point locations of gauging stations.
- `"catchments"`: polygon boundaries of catchments.

Value

An `sf` object.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read gauges
gauges <- read_geospatial("gauges")

# Read catchments
catchments <- read_geospatial("catchments")

# Plot
plot(catchments$geometry)
plot(gauges$geometry, add = TRUE, col = "red")

## End(Not run)
```

`read_metadata` *Read CAMELS-PE Station Metadata*

Description

Reads the station metadata table from the CAMELS-PE dataset. This file contains the main information for each gauging station, such as the station identifier, name, location, drainage area, river, basin, or other available metadata fields depending on the CAMELS-PE dataset version.

Usage

```
read_metadata(path = get_camels_path())
```

Arguments

`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with `set_camels_path()` is used.

Details

The function reads the file:

`01_metadata/stations.csv`

from the CAMELS-PE root directory.

Value

A tibble with CAMELS-PE station metadata.

Examples

```
## Not run:
# Define the CAMELS-PE dataset path
set_camels_path("D:/DATA/CAMELS-PE")

# Read station metadata
```

```
stations <- read_metadata()

# Preview the first rows
head(stations)

## End(Not run)
```

---

read_timeseries *Read CAMELS-PE time series*

Description

Reads time series data from the CAMELS-PE dataset. The function supports two modes:

Usage

```
read_timeseries(
  gauge_id = NULL,
  global = FALSE,
  vars = NULL,
  path = get_camels_path()
)
```

Arguments

`gauge_id`	Optional character vector of gauge IDs. Required if `global = FALSE`.
`global`	Logical. If TRUE, reads the global file. Default is FALSE (recommended for performance).
`vars`	Optional character vector of variables to keep.
`path`	Optional CAMELS-PE root path.

Details

- **Global mode**: reads the full dataset from `timeseries.csv`.
- **By-catchment mode** (recommended): reads only selected catchments from individual files in `by_catchment/`.

Value

A tibble with time series data.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read a single catchment
ts <- read_timeseries(gauge_id = "PE_0001")

# Read multiple catchments
ts <- read_timeseries(gauge_id = c("PE_0001", "PE_0002"))
```

```
# Read global dataset
ts_all <- read_timeseries(global = TRUE)

## End(Not run)
```

`set_camels_path` *Set CAMELS-PE dataset path*

Description

Set CAMELS-PE dataset path

Usage

```
set_camels_path(path)
```

Arguments

`path`	Character. Path to the CAMELS-PE root directory.

Value

Invisibly returns the normalized path.

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